ANALYSIS REPORT

RECURRENCE RISK PREDICTION FOR THYROID CANCER ACCORDING TO ATA 2025 GUIDELINES

Firth Penalized Logistic Regression

Retrospective Study, 114 patients

Study period: 2024

Firth method resolves complete separation in rare mutation groups

1. Study Sample Characteristics

Total patients: 114. Recurrence events: 7 (6.1%).

1.1. ATA 2025 PTC Risk Stratification (4-level)

(ATA 2025 PTC 4-level: Low / Low-Intermediate / Intermediate-High / High)

Risk Level	N	%	Recurrence
Low	69	60.5%	5/69 (7.2%)
Low-Intermediate	2	1.8%	1/2 (50.0%)
Intermediate-High	15	13.2%	0/15 (0.0%)
High	28	24.6%	1/28 (3.6%)
Total	114	100%	7/114 (6.1%)

Note: Recurrence rates do not increase monotonically across levels due to missing histopathological criteria (aggressive histology, margins, ETE, vascular invasion) unavailable in this dataset.

1.2. Descriptive Statistics

Variable	N (n=114)	%
Multifocal (>1 nodule)	31	27.2%
Bilateral involvement	32	28.1%
Tumor size >1cm	21	18.4%
Tumor size >4cm	1	0.9%
T stage T3+	28	24.6%
LN metastasis (N+)	59	51.8%
Lateral neck meta. (N1b)	8	7.0%
Central LN >=5	16	14.0%
Lateral neck LN (+)	21	18.4%
BRAF V600E	45	39.5%
Other BRAF	2	1.8%
TERT mutation	6	5.3%
TP53 mutation	2	1.8%
Any mutation	51	44.7%
Co-mutation (>=2 genes)	4	3.5%
BRAF + TERT	3	2.6%
BRAF + TP53	0	0.0%

2. Univariate Firth Logistic Regression

Firth penalized likelihood logistic regression resolves complete separation in mutation groups with zero events, producing finite odds ratios and confidence intervals for all variables. Variables with <2 positive cases (tumor size >4cm, BRAF+TP53) are excluded.

Variable	N	Events	OR	95% CI	p-value
Multifocal (>1 nodule)	114	7	2.170	0.495 - 9.514	0.304
Bilateral involvement	114	7	2.070	0.473 - 9.060	0.334
Tumor size >1cm	114	7	0.985	0.151 - 6.432	0.987
Tumor size >4cm	114	7	--	--	excluded
T stage T3+	114	7	0.676	0.106 - 4.320	0.679
LN metastasis (N+)	114	7	4.414	0.707 - 27.553	0.112
Lateral neck (N1b)	114	7	3.092	0.409 - 23.381	0.274
Central LN >=5	114	7	2.931	0.574 - 14.980	0.196
Lateral neck LN (+)	114	7	0.985	0.151 - 6.432	0.987
BRAF V600E	114	7	1.199	0.278 - 5.176	0.808
BRAF (any)	114	7	1.110	0.257 - 4.788	0.889
TERT mutation	114	7	1.041	0.043 - 25.413	0.980
TP53 mutation	114	7	2.813	0.063 - 124.823	0.593
Any mutation	114	7	0.954	0.222 - 4.110	0.950
Co-mutation (>=2)	114	7	1.533	0.054 - 43.723	0.803
BRAF + TERT	114	7	1.990	0.060 - 66.025	0.700
BRAF + TP53	114	7	--	--	excluded

*Note: Firth penalization uses Jeffrey's prior penalty (0.5*log|I(beta)|), guaranteeing finite estimates even with complete separation. TERT (OR=1.04, CI=0.04-25.41) and TP53 (OR=2.81, CI=0.06-124.82) now have finite and interpretable estimates, though wide CIs reflect low event count.*

2.1. Key Observations

- N+ is the strongest predictor (OR=4.41; p=0.112), approaching significance.
- Central LN >=5 shows moderate risk (OR=2.93), consistent with ATA guidelines.
- TP53 Firth OR=2.81 suggests possible risk elevation but CI is very wide.
- TERT OR=1.04 indicates no association with recurrence in this sample.
- BRAF V600E OR=1.20, not significant (p=0.81).

3. Multivariate Firth Logistic Regression

Four models fitted with Firth penalized logistic regression.

3.1. Model 1 - Clinical (N=114, events=7)

Variable	OR	95% CI	p-value
Size >1cm	0.716	0.113 - 4.526	0.723
T3+	0.570	0.094 - 3.461	0.541
N+	4.979	0.840 - 29.503	0.077

Hosmer-Lemeshow: chi2=4.07, df=8, p=0.851

3.2. Model 2 - Genetic (N=114, events=7)

Variable	OR	95% CI	p-value
BRAF V600E	1.180	0.282 - 4.934	0.820
TERT	1.040	0.045 - 23.937	0.980

Hosmer-Lemeshow: chi2=7.86, df=8, p=0.447

3.3. Model 3 - Combined (N=114, events=7)

Variable	OR	95% CI	p-value
Size >1cm	0.710	0.119 - 4.218	0.706
T3+	0.545	0.090 - 3.297	0.508
N+	4.650	0.863 - 25.049	0.074
BRAF V600E	1.111	0.265 - 4.661	0.886
TERT	0.978	0.043 - 22.227	0.989

Hosmer-Lemeshow: chi2=6.37, df=8, p=0.606

3.4. Model 4 - N+ and Co-mutation (N=114, events=7)

Variable	OR	95% CI	p-value
N+	4.118	0.710 - 23.891	0.115
Co-mutation	2.163	0.065 - 71.702	0.666

Hosmer-Lemeshow: chi2=4.11, df=8, p=0.848

4. Key Findings and Conclusions

- **Firth Method**
  Firth penalized logistic regression was used to address complete separation caused by rare mutation groups (TERT n=6, TP53 n=2, co-mutation n=4). All estimates now have finite odds ratios and confidence intervals.

- **Recurrence Rate**
  The overall recurrence rate was 6.1% (7/114 patients).

- **N+ as Predictor**
  N+ is the strongest and most consistent predictor: univariate OR=4.41 ($p=0.112$); Model 1 OR=4.98 ($p=0.077$). The consistent trend across all models supports its clinical relevance.

- **TERT and TP53**
  Firth provides finite OR for TERT (1.04) and TP53 (2.81), but wide CIs indicate insufficient data. TP53 OR=2.81 hints at risk elevation, while TERT shows no association.

- **BRAF V600E**
  BRAF V600E (39.5% prevalence) is not associated with recurrence (OR=1.20; $p=0.81$).

- **Model Fit**
  All 4 multivariate models pass Hosmer-Lemeshow goodness-of-fit test ($p>0.05$), indicating adequate model specification.

- **Limitations**
  The limited number of recurrence events (n=7) substantially reduces statistical power. Firth resolves separation but cannot increase effective sample size. Confidence intervals around rare mutation ORs remain very wide. Larger studies with Cox regression are recommended.

Firth logistic regression (Jeffreys prior). Data encoded from Q3 source file and genetic results file. Encoded data file: DATA_MAHOA_ATA2025.xlsx | Analysis file: KET_QUA_PHAN_TICH_ATA2025_FIRTH.txt